

SkyPilot

A voice-guided stargazing navigator that talks you onto the ISS, planets, and satellites.

The Idea

Point your phone at the sky and ask out loud: “Where’s the ISS?”, “What’s that bright dot?”, “When does the next Starlink train pass over Ithaca?” SkyPilot computes where the object is from real orbital data, compares that to where your phone is pointing, and Grok Voice guides you onto it: “Turn left about 30 degrees... now tilt up a little... there.” An on-screen arrow and an audio-only mode make it usable by low-vision users.

Track Fit

Built with Cursor	Entire web app built in Cursor (Agent for scaffolding, rules file, prompt log).
Grok Voice / Imagine	Voice is the core interface. Imagine renders a “what you’d see through a telescope” view.
Real space data	Live satellite orbits (CelesTrak), planet ephemerides, bright-star catalog.
Navigation theme	Literally navigates your gaze across the sky; accessible audio navigation (a prompt from the opening slides).
Also eligible	Big Red, Software, Design, People’s Choice.

Features

MVP (must have by Saturday night)

- Mobile web page reading phone compass heading and tilt.
- Compute altitude/azimuth of the ISS and 5 naked-eye planets for the user’s location and time.
- “Find it” mode: arrow + spoken directions until the phone is on target.
- Voice questions answered by Grok using the computed data.

Stretch

- Upcoming visible-pass list with notifications.
- Haptic “warmer/colder” vibration for eyes-free use.
- Camera AR overlay with labels.
- Grok Imagine telescope view of the selected object.

Tech Stack and Data

- Frontend: React/Next.js (PWA), `DeviceOrientationEvent` for heading and tilt.
- Orbits: CelesTrak GP/TLE data propagated with `satellite.js`.
- Planets: `astronomy-engine` (JS) or JPL Horizons.

- AI: Grok Voice API (conversation), Grok Imagine (images).
- Hosting: Vercel or similar (HTTPS is required for phone sensors).

Limitations and Risks

- **Judging is indoors, in daylight.** Judging runs 9:00 AM–12:30 PM inside PSB. You cannot actually see the ISS or most planets during the demo.
Mitigation: Build a “demo sky” mode that simulates nighttime at the current location; the pointing guidance still works on the real ceiling. Record a real nighttime clip Saturday night as backup.
- **Phone compass accuracy.** Magnetometers are often off by 10–20° indoors, near laptops, or near steel structures.
Mitigation: Calibrate against a known reference (e.g. the Sun or a bright planet), use a generous “on target” tolerance, and be honest about it in the pitch.
- **Browser sensor permissions.** iOS Safari requires an explicit permission prompt and HTTPS for orientation events; Android browsers differ in what “heading” means.
Mitigation: Test on both platforms Friday night; deploy over HTTPS from the start.
- **ISS visibility is limited.** The ISS is only visible when it is sunlit and the observer is in darkness (around dawn and dusk), on a pass over your location.
Mitigation: Show “visible passes” honestly; always allow “point to where it is” even if it isn’t visible.
- **Voice latency.** A round trip of speech to Grok and back can lag behind someone sweeping their phone around.
Mitigation: Generate the turn-by-turn cues locally (short pre-made phrases or the browser speech API) and use Grok only for conversational questions.
- **Originality.** Sky-map apps (SkyView, Stellarium, Star Walk) already exist.
Mitigation: Differentiate on *voice-first*, *accessible navigation* and conversation, not on the star map.
- **Grok API unknowns.** Exact Voice/Imagine API shape, rate limits, and cost are not verified yet.
Mitigation: Confirm at the Friday 9:30 PM Grok/Cursor workshop; get one call working before building on it.

Scorecard

Demo-able	4/5	Memorable, but needs a simulated sky indoors.
Buildable in 24h	4/5	Mostly frontend and well-known libraries.
Navigation theme	5/5	Direct fit, plus accessibility.
Wow factor	4/5	Talking phone pointing at the sky.
Technical risk (5 = riskiest)	2/5	Low to moderate (sensors, latency).

Verdict

Recommended. It has the best balance of theme fit, demo impact, and buildability. The main risk is the indoor daylight demo, which a simulated-sky mode solves.